

PREDICTIVE PARADOX

Explore the paradox within prediction



Problem statement

The Recruitment Task

1. Background & Context

Accurate electricity demand forecasting is critical for power grid stability and efficient energy management. Overestimating demand leads to wasted generation and financial losses, while underestimating demand causes severe grid instability and load shedding.

Your task is to build a robust machine learning pipeline to predict the short-term power demand based on historical consumption, environmental factors, and economic indicators.

2. Objective

Build a predictive machine learning model to forecast the next hour's electricity demand (demand_mw) on the national grid.

3. The Data Provided

dataset- Predictive Paradox Dataset

- PGCB_date_power_demand.xlsx: Hourly demand and generation data. This is your target variable source.
- weather_data.xlsx: Hourly environmental data (temperature, humidity, precipitation).
- economic_full_1.csv: Annual macroeconomic indicators (World Bank data).

4. Pipeline Requirements & Constraints

You are strictly limited to using classical machine learning architectures (e.g., Random Forest, XGBoost, LightGBM, or similar tree-based/linear regressors). Deep learning architectures (LSTMs, Transformers) or autoregressive packages (ARIMA, Prophet) are not permitted.

To succeed, your pipeline must address the following challenges:

A. Data Preparation & Structural Integrity

The provided datasets are unaltered extractions and contain several real-world anomalies:

- Identify and properly resolve duplicate entries or irregular timestamp frequencies in the core demand dataset.
- The raw demand data contains severe undocumented spikes and extreme values. You are expected to implement a mathematically sound anomaly detection and smoothing/removal strategy.
- Devise a logical mechanism to integrate the annual macroeconomic data into your hourly feature set.

B. Feature Representation

Since you are restricted to algorithms that treat observations independently (non-sequential models), you assume full responsibility for establishing the concept of "time" within your dataset.

- You must engineer appropriate features that allow the model to recognize recent historical trends, daily seasonality, and consumption patterns leading up to the target hour.
- (Hint: Consider how an algorithm can "look back" at previous hours without violating the rules of a supervised tabular structure.)

C. Validation Rigor

- **Strict Chronological Separation:** You must ensure zero data leakage from future hours into your training process. Your evaluation must be performed on a chronologically hold-out test set (e.g., all of 2023).
- **Primary Evaluation Metric:** Mean Absolute Percentage Error (MAPE).

5. Deliverables

A. A clean, reproducible Jupyter Notebook or Python executable
Containing your end-to-end pipeline.

B. A markdown/text summary report

Detailing:

- Your rationale for handling missing data and outliers.
- An explanation of the specific temporal features you chose to engineer and why.
- A visualization of your model's Feature Importances to explain the key drivers behind grid power demand.

Expected Flow

- Goal & Target
 - Use the provided datasets to build a short-term forecasting model for the next hour's grid demand.
 - Target to predict: demand_mw at time $t+1$ using information available up to time t .
- Data Understanding
 - Explore PGCB_date_power_demand.xlsx to understand the structure of the demand/generation time series and the meaning of each column.
 - Explore weather_data.xlsx to see what environmental variables are available at an hourly level.
 - Explore economic_full_1.csv to identify set of relevant annual macro-indicators that could influence long-term demand levels.
- Data Preparation
 - Clean and standardize the time axis in the PGCB data, Handle suspicious spikes / extreme values in demand_mw
 - Reshape and tidy data so that data is organized according to chosen indicators and can be joined to the hourly series by calendar year.
- Feature Engineering
 - Encode calendar/time structure: hour of day, day of week, month, weekend/weekday, and other simple time flags.
 - Design historical demand features so the model can “see” recent behavior:
 - Lags: past few hours and relevant daily/weekly offsets.
 - Rolling aggregates: recent averages/variability over suitable time windows.
 - Define the supervised learning target as “next hour's demand” by appropriately shifting the demand series.

- Train/Test Strategy
 - Sort all rows chronologically and choose a future period (e.g., a recent full year) as a strict hold-out test set.
 - Use only past data (before the test period) for model training and any hyperparameter tuning.
 - Ensure that when computing features for each row, no information from future timestamps leaks into the past
- Modeling
 - Select and train one or more classical regression models (e.g., tree-based or linear methods) consistent with the constraints.
 - Use the engineered tabular features as inputs (X) and the next-hour demand as output (y).
- Evaluation & Final Score
 - Evaluate performance on the chronological hold-out test set using Mean Absolute Percentage Error (MAPE) as the primary score.
 - Report the final test MAPE (%) and briefly interpret its magnitude.
- Interpretation & Reporting
 - Examine model feature importance (or an equivalent explanation method) to identify which factors drive demand the most.
 - In a short report or markdown summary, explain:
 - How missing data and anomalies were handled.
 - Which temporal and external features were engineered and why.
 - Key insights from feature importance.

Submission Structure-

Submit a Github Repository With proper Readme and .ipynb Files

Anti-Cheating Regulations

The following rules will be strictly followed during the selection process for entering the club. Anyone found to violate these rules will be disqualified.

1. Submissions found to be completely AI-generated will be discarded.
2. If two people submit nearly identical codes (plagiarism), both will be disqualified from the recruitment process.
3. If someone is found to leak the code for any submission he /she will be disqualified.

We wish you all the best for the recruitment task. Give it your sincere effort, stay confident, and focus on showing your problem-solving approach, creativity, and willingness to learn. The goal is not just to get the perfect answer, but to demonstrate your thinking process and enthusiasm for AI. Take this as an opportunity to challenge yourself, explore new ideas, and enjoy the learning experience.

If you have any queries regarding the task, feel free to contact:
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THANK YOU !

“UNRAVEL THE PREDICTIVE PARADOX.”